higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 4

(12030534)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Keep subsections of questions together.
5. Rule off across the page on completion of each question.
6. ALL work you do NOT want to be marked must be clearly crossed out.
7. Write neatly and legibly.

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QUESTION 1: ROAD DRAINAGE

1.1 Choose a word(s) from the following list that matches each sentence below. Write only the word(s) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

kerb; sand blankets; shoulder; debris; erode; fords; Spanish windlass; geotextile; manhole; oversaturated; run-off; asphalt

1.1.1 The condition in which the earth or soil is being worn away by the action of water and wind

1.1.2 Floating or submerged solid objects carried away by flowing water.

1.1.3 Constructed across streams where there is minimal water flooding.

1.1.4 A road side structure made from pre-cast reinforced concrete that usually placed at the edges of an urban road.

1.1.5 A device consisting of a rope or chain, in which a lever is inserted to gain power by twisting, pulling and tightening wire.

1.1.6 A permeable fabric that has the ability to separate, filter or drain water when used with soil.

1.1.7 A road layer consisting of a thick layer of coarse sand that has the ability to collect and convey seepage water.

1.1.8 A structure that allows access into a storm water drainage system

1.1.9 When something is completely filled with a substance to the extent that nothing more can be absorbed

1.1.10 The part of the road that will provide a place for a vehicle when it breaks down

(10 × 1) (10)

1.2 List THREE examples of road surfacing hazards that can be caused due to badly drained roads. (3)

1.3 Where are longitudinal side drains usually installed? (1)

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1.4 You are required to supervise the placing and building of manholes at regular intervals along a below ground drainage system for a new road.

1.4.1 What is the purposes for placing manholes at regular intervals along a road drainage system? (2)

1.4.2 Name TWO types of manholes in terms of their shape and design that are commonly used. (2)

1.4.3 Name THREE different parts and components of a pre-cast concrete manhole. (3)

1.5 Briefly explain what the following THREE types of sub-surface drainage systems consist off.

1.5.1 Pipe underdrains (2)

1.5.2 Drilled drains (2)

1.5.3 French drains (2)

1.6 What are the THREE main components soil is made from? (3)
[30]

QUESTION 2: ROAD PAVEMENT

2.1 When looking at the advantages of concrete block paving compared to asphalt paving.

State FIVE situations where concrete block paving will be the best to use in terms of cost and performance. (5)

2.2 What is the purpose of grouting sand? (2)

2.3 What is the required compressive strength of a concrete paving block when applied and used for the following?

2.3.1 Airport landing strip.

2.3.2 Light vehicles in resident areas.

2.3.3 Vehicles in commercial or Industrial areas.

2.3.4 Walkways and pedestrian areas.

2.3.5 Driveways and patios.

(5 × 1) (5)

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2.4 State FIVE requirements that bedding sand should comply with to ensure the regularity of the surface level when laying a concrete block paving. (5)

2.5 Differentiate between the following types of materials:

2.5.1 Tar (2)

2.5.2 Bitumen (2)

2.6 Name FOUR different types of tests used to determine the performance and quality levels of bitumen. (4)

2.7 Name THREE types of bitumin binders that are used to bind the aggregates together when producing asphalt. (3)

2.8 State SIX advantages for sealing road surfaces. (6)

2.9 What percentage of bitumen emulsion is needed in the production of a slurry seal? (1)

2.10 Name FOUR types of joints that can be used in concrete pavements. (4)

2.11 Name ONE type of plant that is used in the mixing or manufacturing of asphalt. (1)

[40]

QUESTION 3: ROAD MAINTENANCE AND REHABILITATION

3.1 Name TWO types of non-structural failures that can be found on a road surface. (2)

3.2 Regular cutting of grass in a road reserve forms an essential part of the road environment.

Name TWO advantages of cutting grass inside the road reserve. (2)

3.3 List THREE structures or methods that can be used to control the flow of storm water on a road surface. (3)

3.4 Various types of traffic control systems are used to identify areas where a road is being repaired.

Name TWO traffic control scenarios that can be considered when repairing parts of the road. (2)

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3.5 Differentiate between the following:

3.5.1 Road rehabilitation (2)

3.5.2 Road reconstruction (2)

3.6 When will it be necessary to reseal an asphalt road surface? (1)

3.7 Sealing cracks on the road is very important and it helps to reduce pavement failure.

Discuss TWO basic procedures to be followed when resealing an asphalt road. (2)

3.8 The process of PPC overlays or *"white topping"* can be used as a rehabilitation method on a flexible pavement.
Name TWO methods that this process of PPC overlays can be divided into. (2)

3.9 When will it be necessary for "hot-in-place" recycling of asphalt when having to do the following?

3.9.1 Repaving (2)

3.9.2 Remixing (2)

3.9 3 Heater scarification (2)

3.10 Full depth repair for a concrete road will be necessary where spalling and cracking of the concrete pavement is beyond minor repairs.

Describe the cause of the following types of failures that will require full depth repairs.

3.10.1 Traverse cracking (2)

3.10.2 Shattered slabs and corner breaks (2)

3.10 3 Punch-outs (2)
[30]

TOTAL: 100

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